Master of Data Analytics

University of Niagara Falls Canada

Master of Data Analytics

Assignment 2 – Sentiment Analysis & Book Recommendation Project

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Introduction

This project covers two different but equally important machine-learning problems:
(1) understanding customer sentiment from text reviews, and
(2) recommending books based on user preferences.

The first part focuses on Yelp reviews and tries to predict whether a review is positive or negative. The second part uses the Goodbooks-10K dataset to build a system that suggests books to readers.

Both problems are common in real-world businesses. Companies constantly deal with large amounts of customer feedback and user behavior data. Turning these massive datasets into meaningful insights is exactly the type of work data analysts and data scientists do daily.

Challenge 1–Yelp Sentiment Analysis

1.1 Why this problem matters

Restaurants, delivery services, and online review platforms receive thousands of reviews every day.
Manually reading each review isn't realistic. A reliable sentiment analysis model can help a business:

- Quickly notice unhappy customers
- Identify service issues
- Track improvements or declines over time
- Making decisions backed by data rather than assumptions

The goal here is simple: given the text of a Yelp review, classify it as **positive** or **negative**.

1.2 Understanding the data

The Yelp Polarity dataset was loaded using the load_dataset("yelp_polarity") function. The dataset provides two things:

the review text

a label (0 = negative, 1 = positive)

What I found during EDA:

Balanced dataset: There were almost equal numbers of positive and negative reviews. This is great because it prevents the model from leaning unfairly toward one class.

Review examples: Positive reviews usually talked about good food, polite staff, and quick service. Negative ones often shared issues like delays, rude behavior, or poor food quality.

Review length: Most reviews were short, but there were a few very long ones. This matters because some models handle long text differently.

These observations helped me understand the type of challenges the models may face.

1.3 Baseline Model: TF-IDF + Random Forest

I started with a simple and classic machine-learning setup:

- Convert text to numeric features using **TF-IDF**
- Train a **Random Forest classifier**

This “baseline” model already performed well—around **89% accuracy**. For a simple model, that’s a solid start.
The confusion matrix showed that the model handled both positive and negative reviews fairly evenly.

1.4 Improving the Model with Grid Search

To push performance a bit further, I used **GridSearchCV** to find the best hyperparameters. Since this process takes a lot of time, I tuned the model using a subset of 20,000 reviews.

After trying different numbers of trees, depths, and splitting rules, the optimized model achieved roughly **90% accuracy**.
The improvement wasn’t huge, but it made the model more stable and consistent.

1.5 Transformer Models: DistilBERT and RoBERTa

Once the baseline models were completed, I tested two advanced models from Hugging Face:

1. DistilBERT

A lightweight Transformer model fine-tuned for sentiment classification.
It gave slightly better performance than the optimized Random Forest.

2. RoBERTa (Twitter-based sentiment model)

This model originally outputs **negative**, **neutral**, or **positive**, so I grouped neutral and positive together for this binary task.
RoBERTa outperformed all previous models with about **91–92% accuracy**.

What stood out:

- Transformer models understood subtle expressions much better than TF-IDF.
- They handled sarcasm and natural speech patterns more effectively.
- They reduced misclassification of negative reviews, which is very important for businesses that want to respond quickly to unhappy customers.

1.6 Business Takeaways

From a business perspective:

- Even small accuracy improvements matter when dealing with thousands of reviews daily.
- Transformer-based models give more reliable sentiment predictions, especially for long or emotionally complex reviews.
- However, Random Forests are faster and easier to deploy if resources are limited.
- A company with enough computing power should prefer Transformers, while smaller businesses can still benefit from optimized TF-IDF models.

Challenge 2: Book Recommendation System (Goodbooks-10K)

2.1 Why recommendations are important

Online bookstores and reading platforms depend heavily on recommendation engines.
They help users discover books they might enjoy, increase engagement, and improve user satisfaction.

The Goodbooks dataset includes:

- **~6 million ratings**
- **53k users**
- **10k books**

This is a classic collaborative-filtering scenario.

2.2 What the data looks like

During EDA, I found:

- Most ratings were 4 or 5 stars — people rate books they already like.
- Most users rated very few books, while a few users rated hundreds (long-tail behavior).
- Similarly, some books had thousands of ratings, while many had only a handful.

All of this means the dataset is **extremely sparse**, which is typical for recommendation system data.

2.3 ALS Model (Alternating Least Squares)

I used the **ALS algorithm** from the implicit library to generate recommendations.

What ALS does in simple words:

- It breaks down the giant user-book rating matrix into smaller pieces that capture hidden patterns.
- These hidden patterns represent things like:
  - preference for certain genres
  - favorite authors
  - writing styles a user likes
- ALS learns these patterns even if users have very few ratings.

Hyperparameters used:

- 50 latent factors
- 0.1 regularization
- 15 iterations

These choices balance training speed and prediction quality.

2.4 Example Recommendations

After training the model, I selected a sample user and generated a list of recommended books. The model returned book indices along with confidence scores.

If the book titles are mapped using books.csv, the final list becomes far more meaningful (for example):

1. *The Hobbit* — Tolkien
2. *The Fellowship of the Ring* — Tolkien
3. *Harry Potter and the Goblet of Fire* — J.K. Rowling

These results show that the model understands the user's reading taste and suggests books in a similar space.

2.5 Business Takeaways

- ALS is efficient for large-scale platforms dealing with millions of ratings.
- It does not require detailed content information — only user behavior patterns.
- Such recommendation systems keep users engaged, help them explore new books, and increase retention.
- For businesses, better recommendations mean higher sales and stronger user loyalty.

Conclusion

This project highlights how machine learning can support two major business needs:

- **Understanding customers through sentiment**
- **Helping users discover relevant content through recommendations**

From the Yelp analysis, I observed that Transformer models are the most accurate, and even small improvements can matter at scale.

From the Good books' recommender, ALS proved to be a practical and powerful method for personalized suggestions.

Overall, both challenges demonstrate how data-driven systems can support decision-making, improve customer satisfaction, and provide personalized user experiences.